

TECNOLÓGICA NACIONAL
DE MÉXICO **INSTITUTO**
TECNOLÓGICO DE
NUEVO LEÓN

SUBDIRECCIÓN ACADÉMICA DEPARTAMENTO DE CIENCIAS BÁSICAS

INGENIERÍA EN SISTEMAS COMPUTACIONALES

MATERIA:

REDES DE COMPUTADORAS

TEMA:

REDES DE COMPUTADORA

Nombre de la actividad:

PROYECTO SERVIDOR NACIONAL

DOCENTE:

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SERVIDOR NACIONAL

PERU

Aquí haremos una arquitectura de 3 Niveles para 24 Departamentos

ESTRUCTURA JERÁRQUICA

NIVEL 1: LIMA (Corazón de la red)

- Función: Control central de toda la red
- IP: 10.14.1.0/24

NIVEL 2: 5 SEDES REGIONALES (Jefes de zona)

1. REGIÓN NORTE - LIBERTAD

- Departamentos: Tumbes, Piura, Lambayeque, Cajamarca, Amazonas, La Libertad
- IP: 10.13.0.0/16

2. REGIÓN CENTRO NORTE - HUÁNUCO

- Departamentos: Áncash, Pasco, Junín, Huancavelica, Huánuco
- IP: 10.10.0.0/16

3. REGIÓN CENTRO SUR - HUANUCO

- Departamentos: Ica, Ayacucho, Apurímac, Junín
- IP: 10.12.0.0/16

4. REGIÓN SUR - AREQUIPA

- Departamentos: Moquegua, Tacna, Arequipa
- IP: 10.4.0.0/16

5. REGIÓN ORIENTE - LORETOS

- Departamentos: San Martín, Ucayali, Madre de Dios, Cusco, Puno, Loreto
- IP: 10.16.0.0/16

NIVEL 3: DEPARTAMENTOS (Oficinas locales)

- Cada uno con IP única: 10.[NÚMERO].0.0/16
- Se conectan a su sede regional

POR QUÉ ESTA ESTRUCTURA

- MÁS RÁPIDO: Comunicación local no pasa por Lima
- MÁS ESTABLE: Si falla Lima, regiones siguen trabajando
- MÁS ORGANIZADO: Cada región maneja sus departamentos
- MÁS ESCALABLE: Fácil agregar nuevos departamentos

EQUIPOS POR NIVEL

Nivel 1: Router 2911 + Switch 2960 + Server-PT

Nivel 2: Router 1941 + Switch 2960 + Server-PT

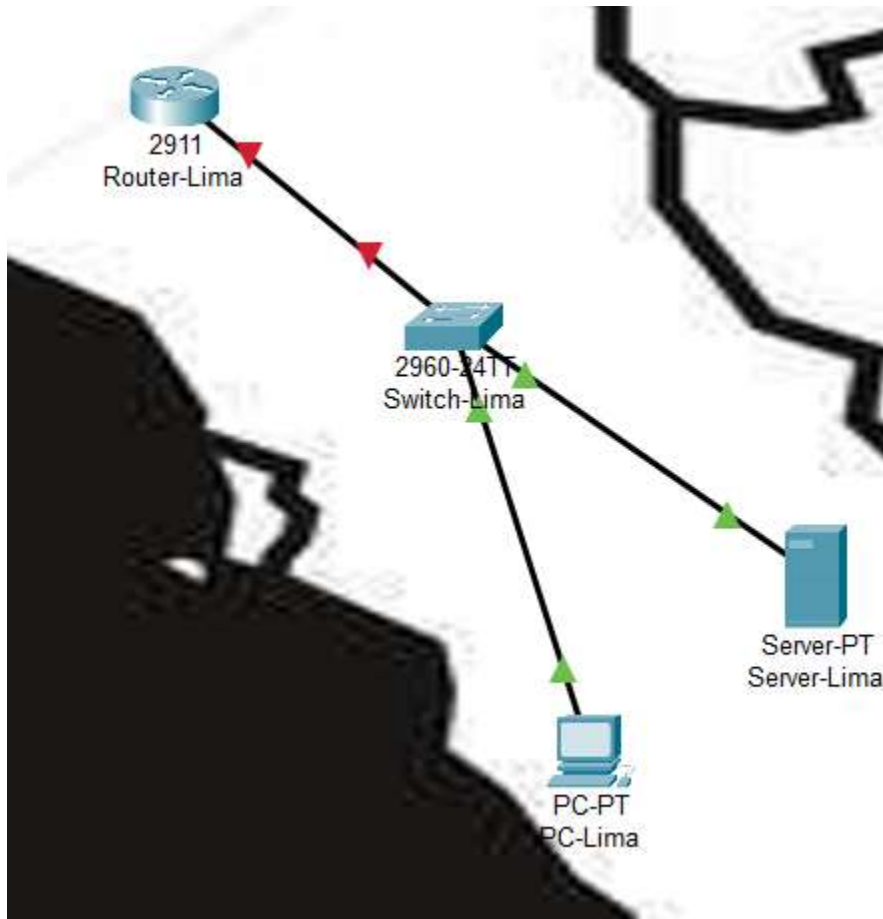
Peru tiene 24 provincias(estados) + 1 que es callao que es autónomo de peru en este caso lo descartaremos y lo tomaremos junto al servidor de lima

1- LIMA

Empezaremos con la capital de peru donde cada servidor tendrá:

- 1 Router 2911
- 1 Switch 2960
- 1 Server-PT
- 1 PC-PT

Y se conectaran con cables **straight-through**.



Configuracion de Router

```
enable
configure terminal

interface gigabitethernet 0/0
ip address 10.14.1.1 255.255.255.0
no shutdown

exit
exit
write memory
```

Configuracion de switch

```
interface range fastethernet 0/1-24
no shutdown

exit
exit
write memory
```

```
enable
configure terminal

ip address 10.14.1.10 255.255.255.0
no shutdown
exit

ip default-gateway 10.14.1.1

exit
write memory
```

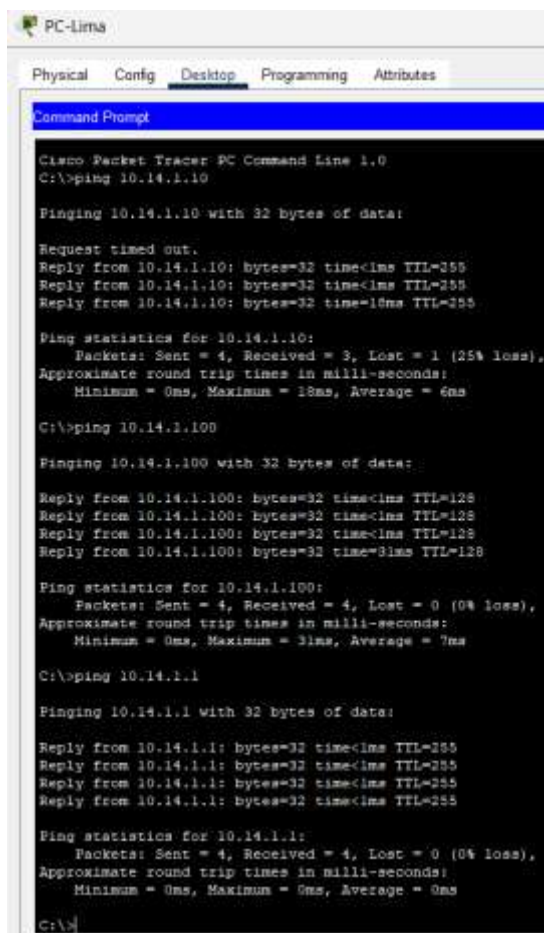
Configuracion del server

- IP Address: 10.14.1.100
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.14.1.1

Configuracion de PC

- IP Address: 10.14.1.50
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.14.1.1

PRUEBAS DE CONECTIVIDAD



```
PC-Lima
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.14.1.10

Pinging 10.14.1.10 with 32 bytes of data:

Request timed out.
Reply from 10.14.1.10: bytes=32 time<1ms TTL=255
Reply from 10.14.1.10: bytes=32 time<1ms TTL=255
Reply from 10.14.1.10: bytes=32 time=18ms TTL=255

Ping statistics for 10.14.1.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 18ms, Average = 6ms

C:\>ping 10.14.1.100

Pinging 10.14.1.100 with 32 bytes of data:

Reply from 10.14.1.100: bytes=32 time<1ms TTL=128
Reply from 10.14.1.100: bytes=32 time<1ms TTL=128
Reply from 10.14.1.100: bytes=32 time<1ms TTL=128
Reply from 10.14.1.100: bytes=32 time=31ms TTL=128

Ping statistics for 10.14.1.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 31ms, Average = 7ms

C:\>ping 10.14.1.1

Pinging 10.14.1.1 with 32 bytes of data:

Reply from 10.14.1.1: bytes=32 time<1ms TTL=255
Reply from 10.14.1.1: bytes=32 time<1ms TTL=255
Reply from 10.14.1.1: bytes=32 time<1ms TTL=255
Reply from 10.14.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.14.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

SERVIDORES REGIONALES

Region Norte

2- Libertad

- 1 Router 2911
- 1 Switch 2960
- 1 Server-PT
- 1 PC-PT

Y se conectaran con cables **straight-through**.

Cable SERIAL (rojo)

- Lima Serial 0/0/0 → Libertad Serial 0/0/0

Configuracion de Router

```
enable
configure terminal

INTERFAZ HACIA LIMA (WAN)
interface serial 0/0/0
ip address 172.16.1.2 255.255.255.252
no shutdown
description ENLACE-A-LIMA

INTERFAZ LOCAL LIBERTAD (LAN)
interface gigabitethernet 0/0
ip address 10.13.1.1 255.255.255.0
no shutdown

ip route 0.0.0.0 0.0.0.0 172.16.1.1

end
write memory
```


Configuracion de Router-Lima

```
enable
configure terminal

interface serial 0/0/0
ip address 172.16.1.1 255.255.255.252
clock rate 64000
no shutdown

ip route 10.13.0.0 255.255.0.0 172.16.1.2

end
write memory
```

Configuracion de switch

```
enable
configure terminal

interface vlan 1
ip address 10.13.1.10 255.255.255.0
no shutdown
exit

ip default-gateway 10.13.1.1

end
write memory
```

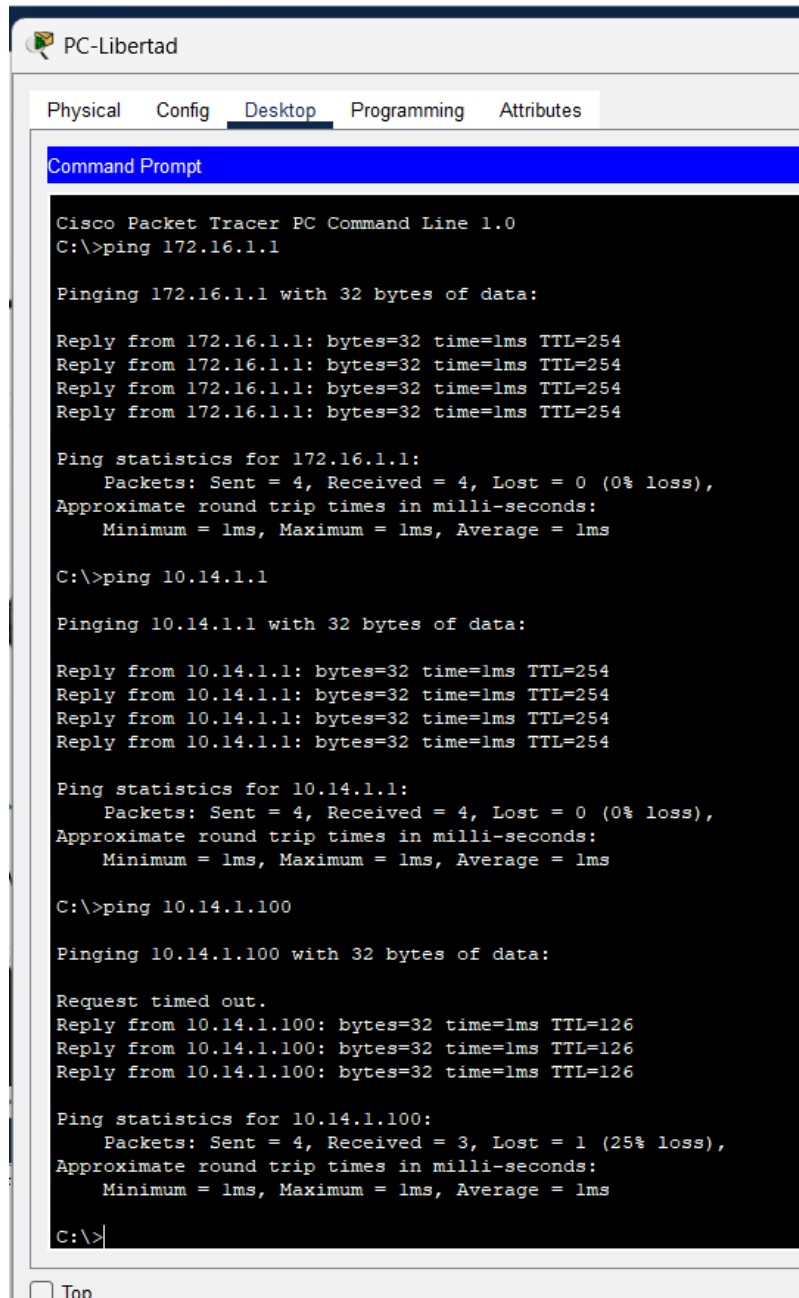
Configuracion del server

- IP Address: 10.13.1.100
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.13.1.1

Configuracion de PC

- IP Address: 10.13.1.50
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.13.1.1

Pruebas de conectividad



The screenshot shows a window titled "PC-Libertad" with tabs for Physical, Config, Desktop, Programming, and Attributes. The "Desktop" tab is active, displaying a "Command Prompt" window. The Command Prompt shows the output of three ping commands executed in a Cisco Packet Tracer PC Command Line 1.0 environment.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.1.1

Pinging 172.16.1.1 with 32 bytes of data:

Reply from 172.16.1.1: bytes=32 time=1ms TTL=254
Reply from 172.16.1.1: bytes=32 time=1ms TTL=254
Reply from 172.16.1.1: bytes=32 time=1ms TTL=254
Reply from 172.16.1.1: bytes=32 time=1ms TTL=254

Ping statistics for 172.16.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping 10.14.1.1

Pinging 10.14.1.1 with 32 bytes of data:

Reply from 10.14.1.1: bytes=32 time=1ms TTL=254
Reply from 10.14.1.1: bytes=32 time=1ms TTL=254
Reply from 10.14.1.1: bytes=32 time=1ms TTL=254
Reply from 10.14.1.1: bytes=32 time=1ms TTL=254

Ping statistics for 10.14.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping 10.14.1.100

Pinging 10.14.1.100 with 32 bytes of data:

Request timed out.
Reply from 10.14.1.100: bytes=32 time=1ms TTL=126
Reply from 10.14.1.100: bytes=32 time=1ms TTL=126
Reply from 10.14.1.100: bytes=32 time=1ms TTL=126

Ping statistics for 10.14.1.100:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>
```

At the bottom left of the window, there is a "Top" button.

Command Prompt

```
Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 172.16.1.2
```

```
Pinging 172.16.1.2 with 32 bytes of data:
```

```
Request timed out.
```

```
Reply from 172.16.1.2: bytes=32 time=1ms TTL=254
```

```
Reply from 172.16.1.2: bytes=32 time=1ms TTL=254
```

```
Reply from 172.16.1.2: bytes=32 time=53ms TTL=254
```

```
Ping statistics for 172.16.1.2:
```

```
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
Minimum = 1ms, Maximum = 53ms, Average = 18ms
```

```
C:\>ping 10.13.1.1
```

```
Pinging 10.13.1.1 with 32 bytes of data:
```

```
Reply from 10.13.1.1: bytes=32 time=1ms TTL=254
```

```
Reply from 10.13.1.1: bytes=32 time=1ms TTL=254
```

```
Reply from 10.13.1.1: bytes=32 time=1ms TTL=254
```

```
Reply from 10.13.1.1: bytes=32 time=1ms TTL=254
```

```
Ping statistics for 10.13.1.1:
```

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

```
C:\>ping 10.13.1.100
```

```
Pinging 10.13.1.100 with 32 bytes of data:
```

```
Request timed out.
```

```
Reply from 10.13.1.100: bytes=32 time=2ms TTL=126
```

```
Reply from 10.13.1.100: bytes=32 time=34ms TTL=126
```

```
Reply from 10.13.1.100: bytes=32 time=1ms TTL=126
```

```
Ping statistics for 10.13.1.100:
```

```
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
Minimum = 1ms, Maximum = 34ms, Average = 12ms
```

3- AMAZONAS

- 1 Router 1941
- 1 Switch 2960
- 1 Server-PT
- 1 PC-PT

Y se conectaran con cables **straight-through**.

Cable SERIAL (rojo)

- Libertad Serial 0/0/1 → Amazonas Serial 0/0/0

Configuracion de Router

```
enable
configure terminal

interface serial 0/0/0
ip address 172.17.1.2 255.255.255.252
no shutdown

interface gigabitethernet 0/0
ip address 10.1.1.1 255.255.255.0
no shutdown

ip route 0.0.0.0 0.0.0.0 172.17.1.1

end
write memory
```

Actualizar router Libertad

```
enable
configure terminal

interface serial 0/0/1
ip address 172.17.1.1 255.255.255.252
clock rate 64000
no shutdown

ip route 10.1.0.0 255.255.0.0 172.17.1.2

end
write memory
```

Configuracion de switch

```
enable
configure terminal

interface vlan 1
ip address 10.1.1.10 255.255.255.0
no shutdown
exit

ip default-gateway 10.1.1.1

end
write memory
```

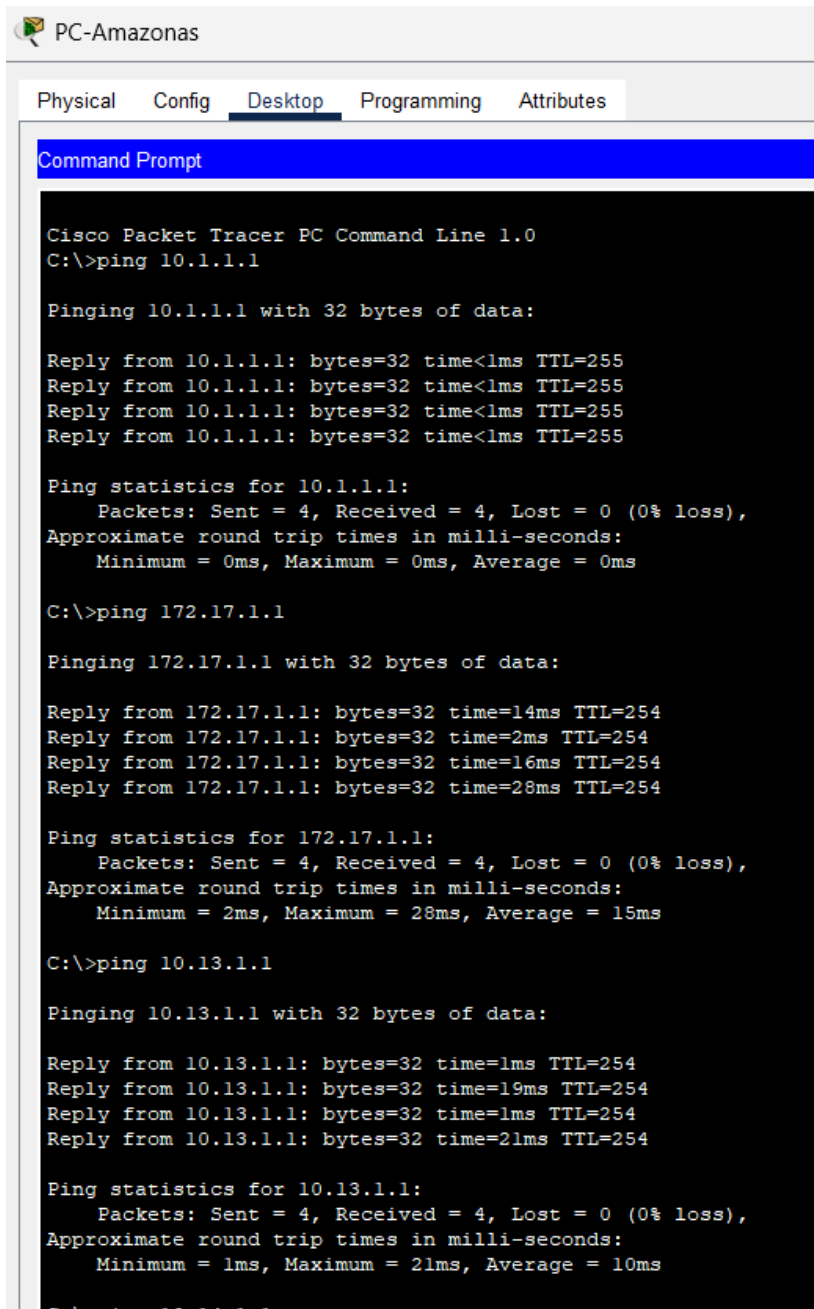
Configuracion del server

- IP Address: 10.1.1.100
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.1.1.1

Configuracion de PC

- IP Address: 10.1.1.50
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.1.1.1

Pruebas de conectividad



PC-Amazonas

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.1.1.1

Pinging 10.1.1.1 with 32 bytes of data:

Reply from 10.1.1.1: bytes=32 time<1ms TTL=255
Reply from 10.1.1.1: bytes=32 time<1ms TTL=255
Reply from 10.1.1.1: bytes=32 time<1ms TTL=255
Reply from 10.1.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.1.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.17.1.1

Pinging 172.17.1.1 with 32 bytes of data:

Reply from 172.17.1.1: bytes=32 time=14ms TTL=254
Reply from 172.17.1.1: bytes=32 time=2ms TTL=254
Reply from 172.17.1.1: bytes=32 time=16ms TTL=254
Reply from 172.17.1.1: bytes=32 time=28ms TTL=254

Ping statistics for 172.17.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 28ms, Average = 15ms

C:\>ping 10.13.1.1

Pinging 10.13.1.1 with 32 bytes of data:

Reply from 10.13.1.1: bytes=32 time=1ms TTL=254
Reply from 10.13.1.1: bytes=32 time=19ms TTL=254
Reply from 10.13.1.1: bytes=32 time=1ms TTL=254
Reply from 10.13.1.1: bytes=32 time=21ms TTL=254

Ping statistics for 10.13.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 21ms, Average = 10ms
```

4- PIURA

- 1 Router 1941
- 1 Switch 2960
- 1 Server-PT
- 1 PC-PT

Y se conectaran con cables **straight-through**.

Cable SERIAL (rojo)

- Libertad Serial 0/1/0 → Piura Serial 0/0/0

Configuracion de Router

```
enable
configure terminal

interface serial 0/0/0
ip address 172.17.2.2 255.255.255.252
no shutdown

interface gigabitethernet 0/0
ip address 10.20.1.1 255.255.255.0
no shutdown

ip route 0.0.0.0 0.0.0.0 172.17.2.1

end
write memory
```

Actualizar router Libertad

```
enable
configure terminal

interface serial 0/1/0
ip address 172.17.2.1 255.255.255.252
clock rate 64000
no shutdown

ip route 10.20.0.0 255.255.0.0 172.17.2.2

end
write memory
```

Configuracion de switch

```
enable
configure terminal

interface vlan 1
ip address 10.20.1.10 255.255.255.0
no shutdown
exit

ip default-gateway 10.20.1.1

end
write memory
```

Actualizar router Lima

```
enable
configure terminal
ip route 10.20.0.0 255.255.0.0 172.16.1.2
end
write memory
```


Configuracion del server

- IP Address: 10.20.1.100
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.20.1.1

Configuracion de PC

- IP Address: 10.20.1.50
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.20.1.1

Pruebas de conectividad

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
```

```
C:\>ping 10.20.1.1
```

```
Pinging 10.20.1.1 with 32 bytes of data:
```

```
Reply from 10.20.1.1: bytes=32 time<1ms TTL=255
```

```
Reply from 10.20.1.1: bytes=32 time<1ms TTL=255
```

```
Reply from 10.20.1.1: bytes=32 time<1ms TTL=255
```

```
Reply from 10.20.1.1: bytes=32 time=14ms TTL=255
```

```
Ping statistics for 10.20.1.1:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 14ms, Average = 3ms
```

```
C:\>ping 10.20.1.10
```

```
Pinging 10.20.1.10 with 32 bytes of data:
```

```
Request timed out.
```

```
Reply from 10.20.1.10: bytes=32 time<1ms TTL=255
```

```
Reply from 10.20.1.10: bytes=32 time<1ms TTL=255
```

```
Reply from 10.20.1.10: bytes=32 time<1ms TTL=255
```

```
Ping statistics for 10.20.1.10:
```

```
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 10.20.1.100
```

```
Pinging 10.20.1.100 with 32 bytes of data:
```

```
Reply from 10.20.1.100: bytes=32 time<1ms TTL=128
```

```
Reply from 10.20.1.100: bytes=32 time<1ms TTL=128
```

```
Reply from 10.20.1.100: bytes=32 time=17ms TTL=128
```

```
Reply from 10.20.1.100: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 10.20.1.100:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 17ms, Average = 4ms
```

Physical Config Desktop Programming Attributes

Command Prompt

```
Packets: Sent = 1, Received = 1, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 17ms, Average = 4ms

C:\>ping 10.13.1.1

Pinging 10.13.1.1 with 32 bytes of data:

Reply from 10.13.1.1: bytes=32 time=1ms TTL=254
Reply from 10.13.1.1: bytes=32 time=1ms TTL=254
Reply from 10.13.1.1: bytes=32 time=1ms TTL=254
Reply from 10.13.1.1: bytes=32 time=1ms TTL=254

Ping statistics for 10.13.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping 10.14.1.1

Pinging 10.14.1.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.14.1.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.14.1.1

Pinging 10.14.1.1 with 32 bytes of data:

Reply from 10.14.1.1: bytes=32 time=36ms TTL=253
Reply from 10.14.1.1: bytes=32 time=41ms TTL=253
Reply from 10.14.1.1: bytes=32 time=2ms TTL=253
Reply from 10.14.1.1: bytes=32 time=37ms TTL=253

Ping statistics for 10.14.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 41ms, Average = 29ms

C:\>
```

5- LAMBAYEQUE

- 1 Router 1941
- 1 Switch 2960
- 1 Server-PT
- 1 PC-PT

Y se conectaran con cables **straight-through**.

Cable SERIAL (rojo)

- Libertad Serial 0/1/1 → Lambayeque Serial 0/0/0

Configuracion de Router

```
enable
configure terminal

interface serial 0/0/0
ip address 172.17.3.2 255.255.255.252
no shutdown

interface gigabitethernet 0/0
ip address 10.17.1.1 255.255.255.0 ! CAMBIADO
no shutdown

ip route 0.0.0.0 0.0.0.0 172.17.3.1

end
write memory
```

Actualizar router Libertad

```
enable
configure terminal

interface serial 0/1/1
ip address 172.17.3.1 255.255.255.252
clock rate 64000
no shutdown
description ENLACE-A-LAMBAYEQUE

ip route 10.17.0.0 255.255.0.0 172.17.3.2

end
write memory
```

Configuracion de switch

```
enable
configure terminal

interface vlan 1
ip address 10.17.1.10 255.255.255.0
no shutdown
exit

ip default-gateway 10.17.1.1

end
write memory
```

Actualizar router Lima

```
enable
configure terminal
ip route 10.17.0.0 255.255.0.0 172.16.1.2
end
write memory
```

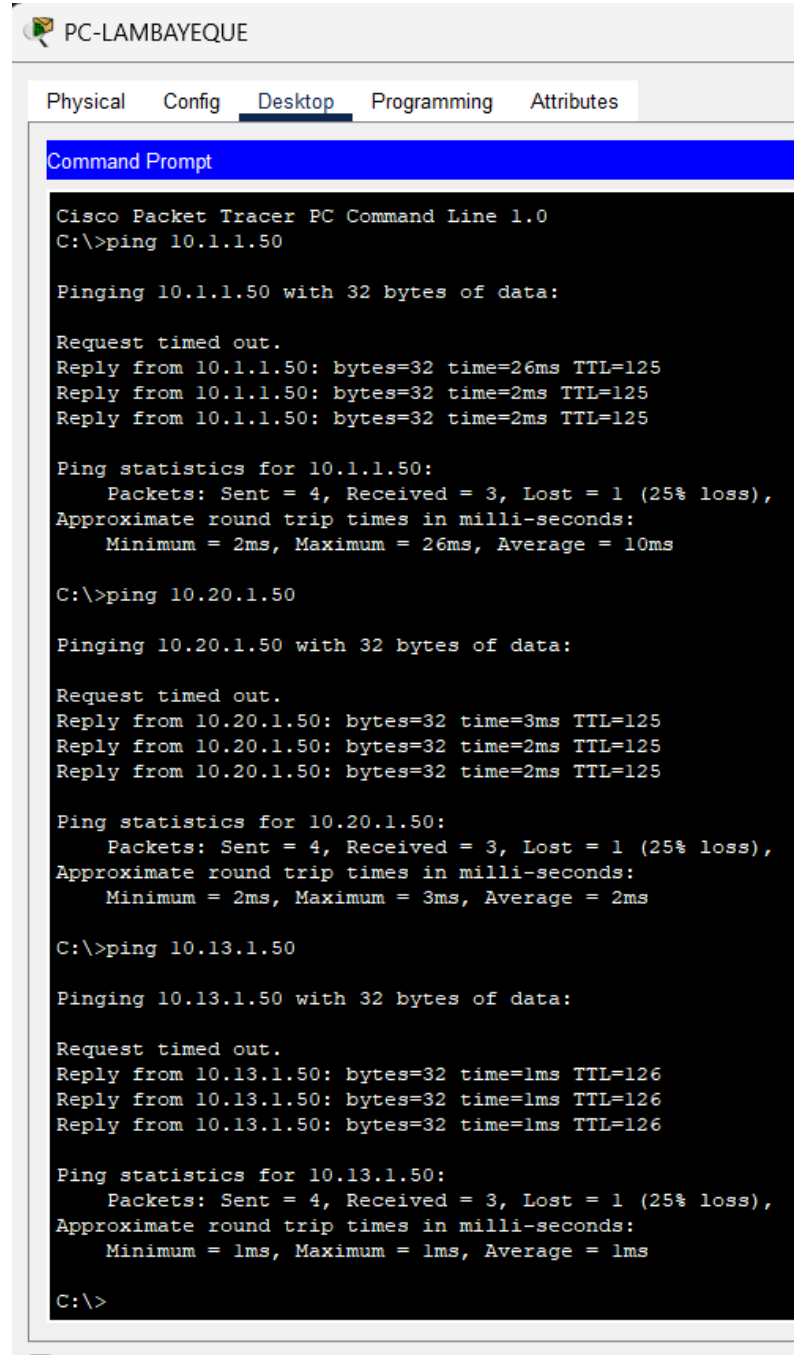
Configuracion del server

- IP Address: 10.17.1.100
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.17.1.1

Configuracion de PC

- IP Address: 10.17.1.50
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.17.1.1

Pruebas de conectividad



The screenshot shows the Cisco Packet Tracer PC Command Line interface for a PC named PC-LAMBAYEQUE. The interface has tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying a Command Prompt window. The Command Prompt shows the execution of three ping commands: ping 10.1.1.50, ping 10.20.1.50, and ping 10.13.1.50. Each command results in a 'Request timed out' message, followed by three successful replies from the respective IP addresses. The ping statistics for each command show 4 packets sent, 3 received, and 1 lost (25% loss). The approximate round trip times in milli-seconds are also displayed for each command.

```
PC-LAMBAYEQUE
Physical  Config  Desktop  Programming  Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.1.1.50

Pinging 10.1.1.50 with 32 bytes of data:

Request timed out.
Reply from 10.1.1.50: bytes=32 time=26ms TTL=125
Reply from 10.1.1.50: bytes=32 time=2ms TTL=125
Reply from 10.1.1.50: bytes=32 time=2ms TTL=125

Ping statistics for 10.1.1.50:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 26ms, Average = 10ms

C:\>ping 10.20.1.50

Pinging 10.20.1.50 with 32 bytes of data:

Request timed out.
Reply from 10.20.1.50: bytes=32 time=3ms TTL=125
Reply from 10.20.1.50: bytes=32 time=2ms TTL=125
Reply from 10.20.1.50: bytes=32 time=2ms TTL=125

Ping statistics for 10.20.1.50:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\>ping 10.13.1.50

Pinging 10.13.1.50 with 32 bytes of data:

Request timed out.
Reply from 10.13.1.50: bytes=32 time=1ms TTL=126
Reply from 10.13.1.50: bytes=32 time=1ms TTL=126
Reply from 10.13.1.50: bytes=32 time=1ms TTL=126

Ping statistics for 10.13.1.50:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>
```

6- Tumbes

- 1 Router 1941
- 1 Switch 2960
- 1 Server-PT
- 1 PC-PT

Y se conectaran con cables **straight-through**.

Cable SERIAL (rojo)

- Libertad Serial 0/1/1 → Lambayeque Serial 0/0/0

Configuracion de Router

```
enable
configure terminal

interface serial 0/0/0
ip address 172.17.4.2 255.255.255.252
no shutdown

interface gigabitethernet 0/0
ip address 10.23.1.1 255.255.255.0
no shutdown

ip route 0.0.0.0 0.0.0.0 172.17.4.1

end
write memory
```


Actualizar router Libertad

```
enable
configure terminal

interface serial 0/2/0
ip address 172.17.4.1 255.255.255.252
clock rate 64000
no shutdown

ip route 10.23.0.0 255.255.0.0 172.17.4.2

end
write memory
```

Configuracion de switch

```
enable
configure terminal

interface vlan 1
ip address 10.23.1.10 255.255.255.0
no shutdown
exit

ip default-gateway 10.23.1.1

end
write memory
```

Actualizar router Lima

```
enable
configure terminal
ip route 10.17.0.0 255.255.0.0 172.16.1.2
end
write memory
```

Configuracion del server

- IP Address: 10.23.1.100
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.23.1.1

Configuracion de PC

- IP Address: 10.23.1.50
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.23.1.100

Pruebas de conectividad

```
C:\>ping 10.17.1.50

Pinging 10.17.1.50 with 32 bytes of data:

Reply from 10.17.1.50: bytes=32 time=2ms TTL=125
Reply from 10.17.1.50: bytes=32 time=2ms TTL=125
Reply from 10.17.1.50: bytes=32 time=2ms TTL=125
Reply from 10.17.1.50: bytes=32 time=2ms TTL=125

Ping statistics for 10.17.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms
```

Command Prompt

Cisco Packet Tracer PC Command Line 1.0

C:\>ping 10.13.1.1

Pinging 10.13.1.1 with 32 bytes of data:

Reply from 10.13.1.1: bytes=32 time=1ms TTL=254

Reply from 10.13.1.1: bytes=32 time=1ms TTL=254

Reply from 10.13.1.1: bytes=32 time=1ms TTL=254

Reply from 10.13.1.1: bytes=32 time=1ms TTL=254

Ping statistics for 10.13.1.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:

Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping 10.1.1.50

Pinging 10.1.1.50 with 32 bytes of data:

Reply from 10.1.1.50: bytes=32 time=2ms TTL=125

Reply from 10.1.1.50: bytes=32 time=2ms TTL=125

Reply from 10.1.1.50: bytes=32 time=66ms TTL=125

Reply from 10.1.1.50: bytes=32 time=3ms TTL=125

Ping statistics for 10.1.1.50:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:

Minimum = 2ms, Maximum = 66ms, Average = 18ms

C:\>ping 10.20.1.50

Pinging 10.20.1.50 with 32 bytes of data:

Reply from 10.20.1.50: bytes=32 time=2ms TTL=125

Reply from 10.20.1.50: bytes=32 time=2ms TTL=125

Reply from 10.20.1.50: bytes=32 time=2ms TTL=125

Reply from 10.20.1.50: bytes=32 time=2ms TTL=125

Ping statistics for 10.20.1.50:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:

Minimum = 2ms, Maximum = 2ms, Average = 2ms

7- Cajamarca

- 1 Router 1941
- 1 Switch 2960
- 1 Server-PT
- 1 PC-PT

Y se conectaran con cables **straight-through**.

Cable SERIAL (rojo)

- Libertad Serial 0/2/1 → Cajamarca Serial 0/0/0

Configuracion de Router

```
enable
configure terminal

interface serial 0/0/0
ip address 172.17.5.2 255.255.255.252
no shutdown

interface gigabitethernet 0/0
ip address 10.6.1.1 255.255.255.0
no shutdown

ip route 0.0.0.0 0.0.0.0 172.17.5.1

end
write memory
```

Actualizar router Libertad

```
enable
configure terminal

interface serial 0/2/1
ip address 172.17.5.1 255.255.255.252
clock rate 64000
no shutdown

ip route 10.6.0.0 255.255.0.0 172.17.5.2

end
write memory
```

Configuracion de switch

```
enable
configure terminal
hostname SW-CAJAMARCA

interface vlan 1
ip address 10.6.1.10 255.255.255.0
no shutdown
exit

ip default-gateway 10.6.1.1

end
write memory
```

Actualizar router Lima

```
enable
configure terminal
ip route 10.6.0.0 255.255.0.0 172.16.1.2
end
write memory
```

Configuracion del server

- IP Address: 10.6.1.100
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.6.1.1

Configuracion de PC

- IP Address: 10.6.1.50
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.6.1.1

Pruebas de conectividad

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
```

```
C:\>ping 10.6.1.1
```

```
Pinging 10.6.1.1 with 32 bytes of data:
```

```
Reply from 10.6.1.1: bytes=32 time<1ms TTL=255
```

```
Reply from 10.6.1.1: bytes=32 time<1ms TTL=255
```

```
Reply from 10.6.1.1: bytes=32 time<1ms TTL=255
```

```
Reply from 10.6.1.1: bytes=32 time<1ms TTL=255
```

```
Ping statistics for 10.6.1.1:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 10.13.1.50
```

```
Pinging 10.13.1.50 with 32 bytes of data:
```

```
Reply from 10.13.1.50: bytes=32 time=1ms TTL=126
```

```
Reply from 10.13.1.50: bytes=32 time=27ms TTL=126
```

```
Reply from 10.13.1.50: bytes=32 time=2ms TTL=126
```

```
Reply from 10.13.1.50: bytes=32 time=1ms TTL=126
```

```
Ping statistics for 10.13.1.50:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 1ms, Maximum = 27ms, Average = 7ms
```

```
C:\>ping 10.1.1.50
```

```
Pinging 10.1.1.50 with 32 bytes of data:
```

```
Reply from 10.1.1.50: bytes=32 time=2ms TTL=125
```

```
Reply from 10.1.1.50: bytes=32 time=3ms TTL=125
```

```
Reply from 10.1.1.50: bytes=32 time=2ms TTL=125
```

```
Reply from 10.1.1.50: bytes=32 time=2ms TTL=125
```

```
Ping statistics for 10.1.1.50:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 2ms, Maximum = 3ms, Average = 2ms
```

```
C:\>ping 10.20.1.50
```

Command Prompt

```
C:\>ping 10.20.1.50
```

```
Pinging 10.20.1.50 with 32 bytes of data:
```

```
Reply from 10.20.1.50: bytes=32 time=2ms TTL=125
Reply from 10.20.1.50: bytes=32 time=2ms TTL=125
Reply from 10.20.1.50: bytes=32 time=37ms TTL=125
Reply from 10.20.1.50: bytes=32 time=2ms TTL=125
```

```
Ping statistics for 10.20.1.50:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 37ms, Average = 10ms
```

```
C:\>ping 10.17.1.50
```

```
Pinging 10.17.1.50 with 32 bytes of data:
```

```
Reply from 10.17.1.50: bytes=32 time=2ms TTL=125
Reply from 10.17.1.50: bytes=32 time=21ms TTL=125
Reply from 10.17.1.50: bytes=32 time=2ms TTL=125
Reply from 10.17.1.50: bytes=32 time=2ms TTL=125
```

```
Ping statistics for 10.17.1.50:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 21ms, Average = 6ms
```

```
C:\>ping 10.23.1.50
```

```
Pinging 10.23.1.50 with 32 bytes of data:
```

```
Reply from 10.23.1.50: bytes=32 time=2ms TTL=125
Reply from 10.23.1.50: bytes=32 time=2ms TTL=125
Reply from 10.23.1.50: bytes=32 time=4ms TTL=125
Reply from 10.23.1.50: bytes=32 time=2ms TTL=125
```

```
Ping statistics for 10.23.1.50:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 4ms, Average = 2ms
```

```
C:\>|
```


SERVIDORES REGIONALES

Region Sur

8- Arequipa

- 1 Router 2911
- 1 Switch 2960
- 1 Server-PT
- 1 PC-PT

Y se conectaran con cables **straight-through**.

Cable SERIAL (rojo)

- Lima Serial 0/0/1 → Arequipa Serial 0/0/0

Configuracion de Router

```
enable
configure terminal

interface serial 0/0/0
ip address 172.16.2.2 255.255.255.252
no shutdown
description ENLACE-A-LIMA

interface gigabitethernet 0/0
ip address 10.4.1.1 255.255.255.0
no shutdown

ip route 0.0.0.0 0.0.0.0 172.16.2.1

end
write memory
```

Actualizar router Lima

```
enable
configure terminal

interface serial 0/0/1
ip address 172.16.2.1 255.255.255.252
clock rate 64000
no shutdown

ip route 10.4.0.0 255.255.0.0 172.16.2.2

end
write memory
```

Configuracion de switch

```
enable
configure terminal

interface vlan 1
ip address 10.4.1.10 255.255.255.0
no shutdown
exit

ip default-gateway 10.4.1.1

end
write memory
```

ya para actualizar la ruta en la capital y en la seccion del norte sabemos el proceso así que nos saltaremos este paso para no abarcar mas espacio

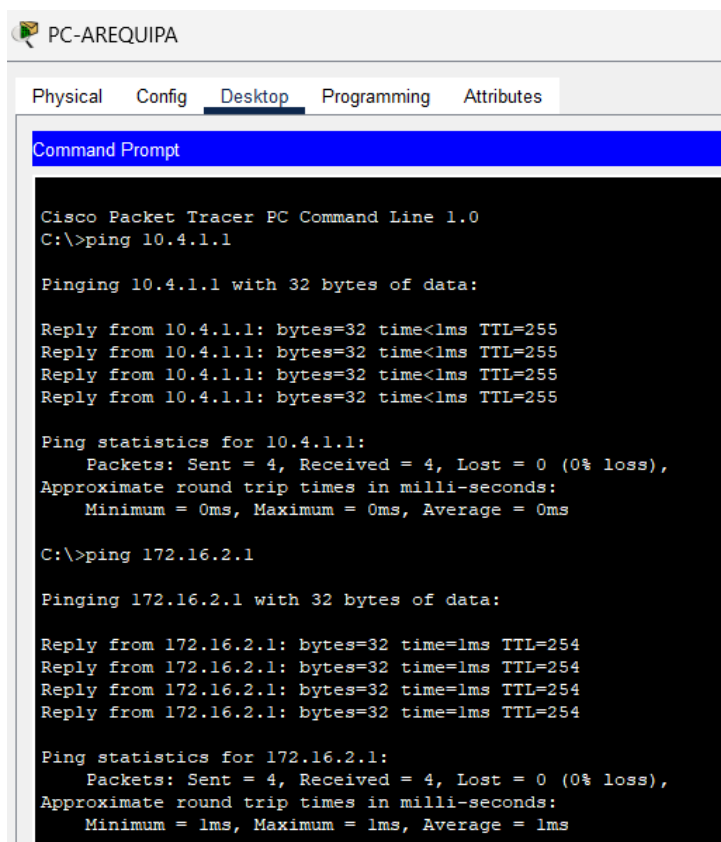
Configuracion del server

- IP Address: 10.4.1.100
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.4.1.1

Configuracion de PC

- IP Address: 10.4.1.50
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.4.1.1

Pruebas de conectividad



The screenshot shows the PC-Arequipa interface with the 'Desktop' tab selected. A Command Prompt window is open, displaying the results of two ping tests. The first test is to 10.4.1.1, showing four successful replies with 32 bytes of data, a time of less than 1ms, and a TTL of 255. The second test is to 172.16.2.1, also showing four successful replies with 32 bytes of data, a time of less than 1ms, and a TTL of 254. Both tests show 0% loss and approximate round trip times in milliseconds.

```
PC-Arequipa
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.4.1.1

Pinging 10.4.1.1 with 32 bytes of data:

Reply from 10.4.1.1: bytes=32 time<1ms TTL=255
Reply from 10.4.1.1: bytes=32 time<1ms TTL=255
Reply from 10.4.1.1: bytes=32 time<1ms TTL=255
Reply from 10.4.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.4.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.16.2.1

Pinging 172.16.2.1 with 32 bytes of data:

Reply from 172.16.2.1: bytes=32 time=1ms TTL=254
Reply from 172.16.2.1: bytes=32 time=1ms TTL=254
Reply from 172.16.2.1: bytes=32 time=1ms TTL=254
Reply from 172.16.2.1: bytes=32 time=1ms TTL=254

Ping statistics for 172.16.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

```
C:\>ping 10.1.1.50
```

```
Pinging 10.1.1.50 with 32 bytes of data:
```

```
Reply from 10.1.1.50: bytes=32 time=10ms TTL=124
```

```
Reply from 10.1.1.50: bytes=32 time=4ms TTL=124
```

```
Reply from 10.1.1.50: bytes=32 time=4ms TTL=124
```

```
Reply from 10.1.1.50: bytes=32 time=5ms TTL=124
```

```
Ping statistics for 10.1.1.50:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 4ms, Maximum = 10ms, Average = 5ms
```

```
C:\>
```

9- Moquegua

- 1 Router 1941
- 1 Switch 2960
- 1 Server-PT
- 1 PC-PT

Y se conectaran con cables **straight-through**.

Cable SERIAL (rojo)

- Arequipa Serial 0/0/2 → Moquegua Serial 0/0/0

Configuracion de Router

```
enable
configure terminal

interface serial 0/0/0
ip address 172.18.1.2 255.255.255.252
no shutdown

interface gigabitethernet 0/0
ip address 10.18.1.1 255.255.255.0
no shutdown
description RED-MOQUEGUA

ip route 0.0.0.0 0.0.0.0 172.18.1.1

end
write memory
```

Actualizar router Arequipa

```
enable
configure terminal

interface serial 0/0/1
ip address 172.18.1.1 255.255.255.252
clock rate 64000
no shutdown

ip route 10.18.0.0 255.255.0.0 172.18.1.2

end
write memory
```

Configuracion de switch

```
enable
configure terminal

interface vlan 1
ip address 10.18.1.10 255.255.255.0
no shutdown
exit

ip default-gateway 10.18.1.1

end
write memory
```

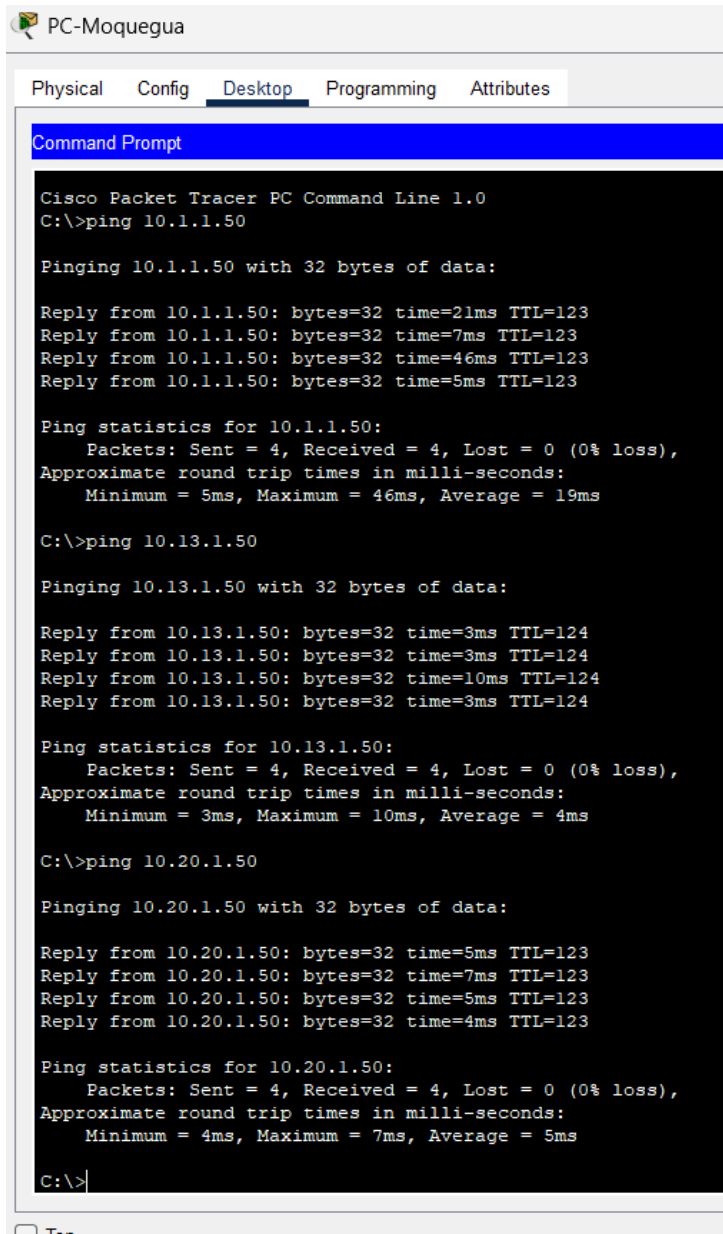
Configuracion del server

- IP Address: 10.4.1.100
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.4.1.1

Configuracion de PC

- IP Address: 10.4.1.50
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.4.1.1

Pruebas de conectividad



PC-Moquegua

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.1.1.50

Pinging 10.1.1.50 with 32 bytes of data:

Reply from 10.1.1.50: bytes=32 time=21ms TTL=123
Reply from 10.1.1.50: bytes=32 time=7ms TTL=123
Reply from 10.1.1.50: bytes=32 time=46ms TTL=123
Reply from 10.1.1.50: bytes=32 time=5ms TTL=123

Ping statistics for 10.1.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 46ms, Average = 19ms

C:\>ping 10.13.1.50

Pinging 10.13.1.50 with 32 bytes of data:

Reply from 10.13.1.50: bytes=32 time=3ms TTL=124
Reply from 10.13.1.50: bytes=32 time=3ms TTL=124
Reply from 10.13.1.50: bytes=32 time=10ms TTL=124
Reply from 10.13.1.50: bytes=32 time=3ms TTL=124

Ping statistics for 10.13.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 10ms, Average = 4ms

C:\>ping 10.20.1.50

Pinging 10.20.1.50 with 32 bytes of data:

Reply from 10.20.1.50: bytes=32 time=5ms TTL=123
Reply from 10.20.1.50: bytes=32 time=7ms TTL=123
Reply from 10.20.1.50: bytes=32 time=5ms TTL=123
Reply from 10.20.1.50: bytes=32 time=4ms TTL=123

Ping statistics for 10.20.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 7ms, Average = 5ms

C:\>
```

Top

10- Tacna

- 1 Router 1941
- 1 Switch 2960
- 1 Server-PT
- 1 PC-PT

Y se conectaran con cables **straight-through**.

Cable SERIAL (rojo)

- Arequipa Serial 0/0/2 → Moquegua Serial 0/0/0

Configuracion de Router

```
enable
configure terminal

interface serial 0/0/0
ip address 172.18.2.2 255.255.255.252
no shutdown

interface gigabitethernet 0/0
ip address 10.22.1.1 255.255.255.0
no shutdown

ip route 0.0.0.0 0.0.0.0 172.18.2.1

end
write memory
```

Actualizar router Arequipa


```
enable
configure terminal

interface serial 0/2/0
ip address 172.18.2.1 255.255.255.252
clock rate 64000
no shutdown

ip route 10.22.0.0 255.255.0.0 172.18.2.2

end
write memory
```

Configuracion de switch

```
enable
configure terminal

interface vlan 1
ip address 10.22.1.10 255.255.255.0
no shutdown
exit

ip default-gateway 10.22.1.1

end
write memory
```

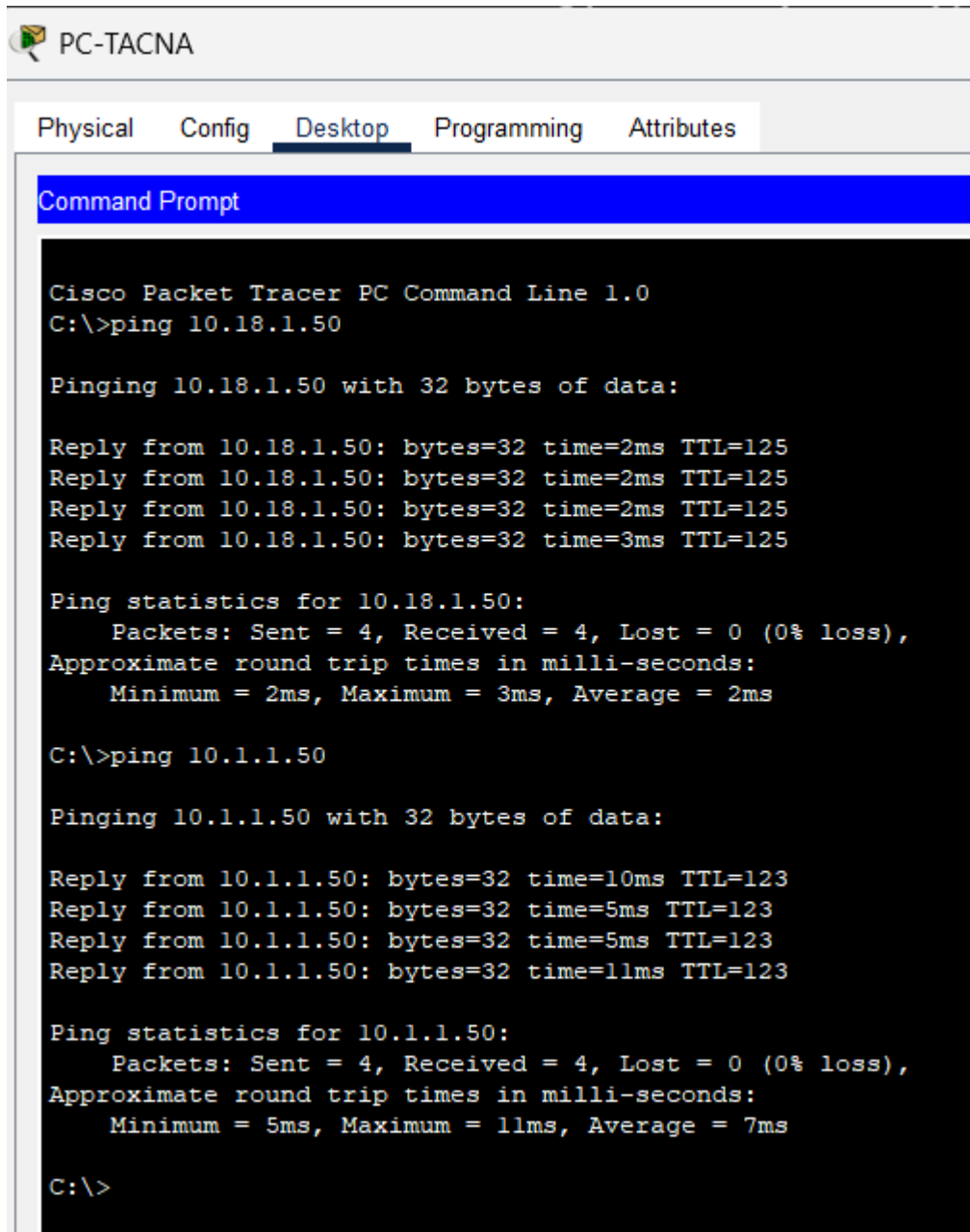
Configuracion del server

- IP Address: 10.22.1.100
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.22.1.1

Configuracion de PC

- IP Address: 10.22.1.50
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.22.1.1

Pruebas de conectividad



The screenshot shows the PC-TACNA interface with the 'Desktop' tab selected. A Command Prompt window is open, displaying the results of two ping tests. The first test is to 10.18.1.50, and the second is to 10.1.1.50. Both tests show successful connectivity with 0% loss.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.18.1.50

Pinging 10.18.1.50 with 32 bytes of data:

Reply from 10.18.1.50: bytes=32 time=2ms TTL=125
Reply from 10.18.1.50: bytes=32 time=2ms TTL=125
Reply from 10.18.1.50: bytes=32 time=2ms TTL=125
Reply from 10.18.1.50: bytes=32 time=3ms TTL=125

Ping statistics for 10.18.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\>ping 10.1.1.50

Pinging 10.1.1.50 with 32 bytes of data:

Reply from 10.1.1.50: bytes=32 time=10ms TTL=123
Reply from 10.1.1.50: bytes=32 time=5ms TTL=123
Reply from 10.1.1.50: bytes=32 time=5ms TTL=123
Reply from 10.1.1.50: bytes=32 time=11ms TTL=123

Ping statistics for 10.1.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 11ms, Average = 7ms

C:\>
```

11- Ica

- 1 Router 1941
- 1 Switch 2960
- 1 Server-PT
- 1 PC-PT

Y se conectaran con cables **straight-through**.

Cable SERIAL (rojo)

- **Arequipa** Serial 0/1/0 → **Ica** Serial 0/0/0

Configuracion de Router

```
enable
configure terminal

interface serial 0/0/0
ip address 172.18.3.2 255.255.255.252
no shutdown

interface gigabitethernet 0/0
ip address 10.11.1.1 255.255.255.0
no shutdown

ip route 0.0.0.0 0.0.0.0 172.18.3.1

end
write memory
```

Actualizar router Arequipa

```
enable
configure terminal

interface serial 0/1/0
ip address 172.18.3.1 255.255.255.252
clock rate 64000
no shutdown

ip route 10.11.0.0 255.255.0.0 172.18.3.2

end
write memory
```

Configuracion de switch

```
enable
configure terminal

interface vlan 1
ip address 10.11.1.10 255.255.255.0
no shutdown
exit

ip default-gateway 10.11.1.1

end
write memory
```

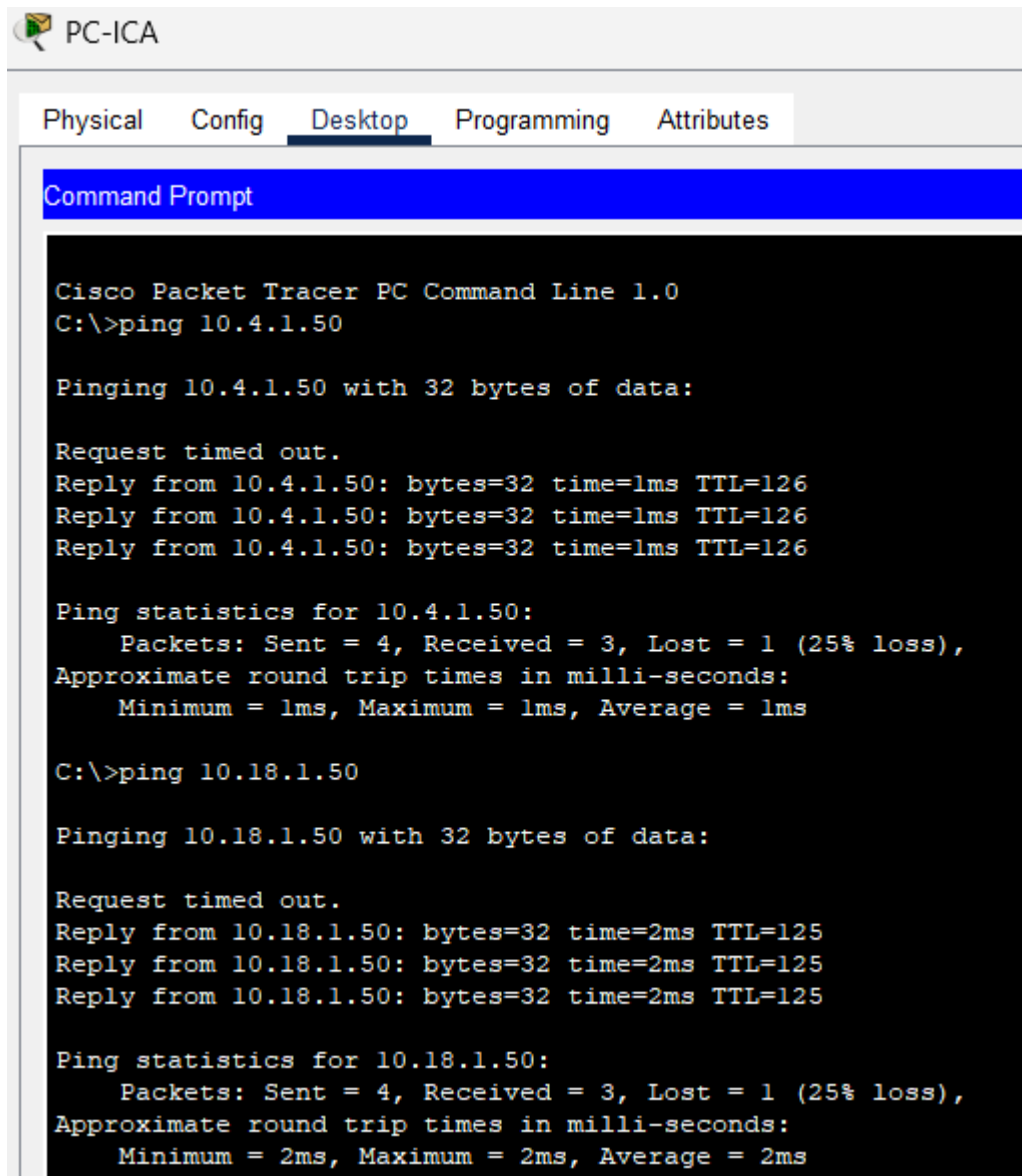
Configuracion del server

- IP Address: 10.11.1.100
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.11.1.1

Configuracion de PC

- IP Address: 10.11.1.50
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.11.1.1

Pruebas de conectividad



The screenshot shows the PC-ICA interface with the 'Desktop' tab selected. A Command Prompt window is open, displaying the results of two ping tests. The first test is to 10.4.1.50, showing a 25% loss of packets. The second test is to 10.18.1.50, also showing a 25% loss of packets. The output for both tests is as follows:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.4.1.50

Pinging 10.4.1.50 with 32 bytes of data:

Request timed out.
Reply from 10.4.1.50: bytes=32 time=1ms TTL=126
Reply from 10.4.1.50: bytes=32 time=1ms TTL=126
Reply from 10.4.1.50: bytes=32 time=1ms TTL=126

Ping statistics for 10.4.1.50:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping 10.18.1.50

Pinging 10.18.1.50 with 32 bytes of data:

Request timed out.
Reply from 10.18.1.50: bytes=32 time=2ms TTL=125
Reply from 10.18.1.50: bytes=32 time=2ms TTL=125
Reply from 10.18.1.50: bytes=32 time=2ms TTL=125

Ping statistics for 10.18.1.50:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 2ms, Average = 2ms
```

12- Ayacucho

- 1 Router 1941
- 1 Switch 2960
- 1 Server-PT
- 1 PC-PT

Y se conectaran con cables **straight-through**.

Cable SERIAL (rojo)

- **Arequipa** Serial 0/1/1 → **Ayacucho** Serial 0/0/0

Configuracion de Router

```
enable
configure terminal

interface serial 0/0/0
ip address 172.18.4.2 255.255.255.252
no shutdown

interface gigabitethernet 0/0
ip address 10.5.1.1 255.255.255.0
no shutdown

ip route 0.0.0.0 0.0.0.0 172.18.4.1

end
write memory
```

Actualizar router Arequipa

```
enable
configure terminal

interface serial 0/1/1
ip address 172.18.4.1 255.255.255.252
clock rate 64000
no shutdown

ip route 10.5.0.0 255.255.0.0 172.18.4.2

end
write memory
```

Configuracion de switch

```
enable
configure terminal

interface vlan 1
ip address 10.5.1.10 255.255.255.0
no shutdown
exit

ip default-gateway 10.5.1.1

end
write memory
```

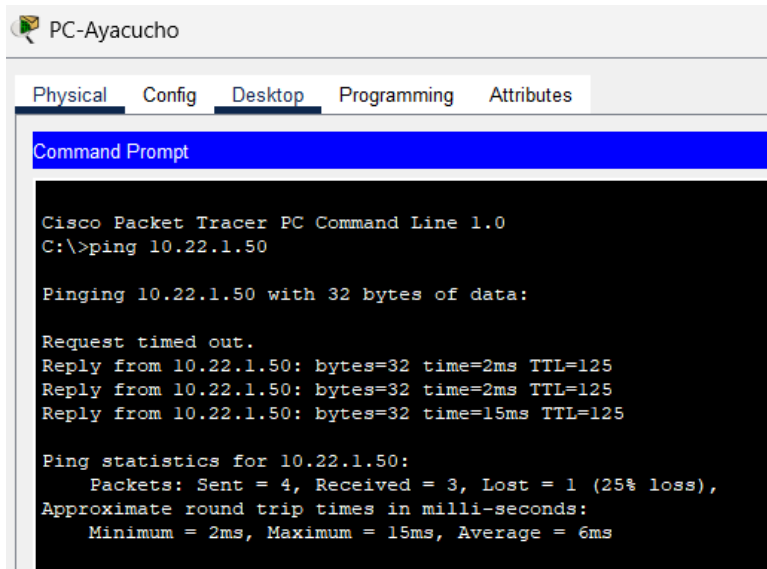
Configuracion del server

- IP Address: 10.5.1.100
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.5.1.1

Configuracion de PC

- IP Address: 10.5.1.50
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.5.1.1

Pruebas de conectividad



PC-Ayacucho

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.22.1.50

Pinging 10.22.1.50 with 32 bytes of data:

Request timed out.
Reply from 10.22.1.50: bytes=32 time=2ms TTL=125
Reply from 10.22.1.50: bytes=32 time=2ms TTL=125
Reply from 10.22.1.50: bytes=32 time=15ms TTL=125

Ping statistics for 10.22.1.50:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 15ms, Average = 6ms
```

```
C:\>ping 10.10.1.50

Pinging 10.10.1.50 with 32 bytes of data:

Reply from 10.10.1.50: bytes=32 time=3ms TTL=124
Reply from 10.10.1.50: bytes=32 time=3ms TTL=124
Reply from 10.10.1.50: bytes=32 time=3ms TTL=124
Reply from 10.10.1.50: bytes=32 time=6ms TTL=124

Ping statistics for 10.10.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 6ms, Average = 3ms

C:\>ping 10.1.1.50

Pinging 10.1.1.50 with 32 bytes of data:

Reply from 10.1.1.50: bytes=32 time=4ms TTL=123
Reply from 10.1.1.50: bytes=32 time=10ms TTL=123
Reply from 10.1.1.50: bytes=32 time=30ms TTL=123
Reply from 10.1.1.50: bytes=32 time=6ms TTL=123

Ping statistics for 10.1.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 30ms, Average = 12ms

C:\>
```


13- Apurimac

- 1 Router 1941
- 1 Switch 2960
- 1 Server-PT
- 1 PC-PT

Y se conectarán con cables **straight-through**.

Cable SERIAL (rojo)

- **Arequipa** Serial 0/1/2 → **Apurimac** Serial 0/0/0

Configuracion de Router

```
enable
configure terminal
interface serial 0/0/0
ip address 172.18.5.2 255.255.255.252
no shutdown
interface gigabitethernet 0/0
ip address 10.3.1.1 255.255.255.0
no shutdown
ip route 0.0.0.0 0.0.0.0 172.18.5.1
end
write memory
```

Actualizar router Arequipa

```
enable
configure terminal
interface serial 0/1/2
ip address 172.18.5.1 255.255.255.252
clock rate 64000
no shutdown
ip route 10.3.0.0 255.255.0.0 172.18.5.2
end
write memory
```

Configuracion de switch

```
enable
configure terminal
interface vlan 1
ip address 10.3.1.10 255.255.255.0
no shutdown
exit
ip default-gateway 10.3.1.1
end
write memory
```

Configuracion del server

- IP Address: 10.3.1.100
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.3.1.1

Configuracion de PC

- IP Address: 10.3.1.50
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.3.1.1

Pruebas de conectividad

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.4.1.100

Pinging 10.4.1.100 with 32 bytes of data:

Request timed out.
Reply from 10.4.1.100: bytes=32 time=4ms TTL=122
Reply from 10.4.1.100: bytes=32 time=4ms TTL=122
Reply from 10.4.1.100: bytes=32 time=11ms TTL=122

Ping statistics for 10.4.1.100:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 11ms, Average = 6ms
```

SERVIDORES REGIONALES

Region Centro-Norte

14- Huanuco

- 1 Router 2911
- 1 Switch 2960
- 1 Server-PT
- 1 PC-PT

Y se conectaran con cables **straight-through**.

Cable SERIAL (rojo)

- Lima Serial 0/0/2 → Huanaco Serial 0/0/0

Configuracion de Router

```
enable
configure terminal

interface serial 0/0/0
ip address 172.16.3.2 255.255.255.252
no shutdown

interface gigabitethernet 0/0
ip address 10.10.1.1 255.255.255.0
no shutdown

ip route 0.0.0.0 0.0.0.0 172.16.3.1

end
write memory
```

Actualizar router Lima

```
enable
configure terminal

interface serial 0/1/0
ip address 172.16.3.1 255.255.255.252
clock rate 64000
no shutdown

ip route 10.10.0.0 255.255.0.0 172.16.3.2

end
write memory
```

Configuracion de switch

```
enable
configure terminal

interface vlan 1
ip address 10.10.1.10 255.255.255.0
no shutdown
exit

ip default-gateway 10.10.1.1

end
write memory
```

Configuracion del server

- IP Address: 10.10.1.100
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.10.1.1

Configuracion de PC

- IP Address: 10.10.1.50
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.10.1.1

Pruebas de conectividad

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.14.1.50

Pinging 10.14.1.50 with 32 bytes of data:

Request timed out.
Reply from 10.14.1.50: bytes=32 time=2ms TTL=126
Reply from 10.14.1.50: bytes=32 time=1ms TTL=126
Reply from 10.14.1.50: bytes=32 time=1ms TTL=126

Ping statistics for 10.14.1.50:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>ping 10.1.1.50

Pinging 10.1.1.50 with 32 bytes of data:

Reply from 10.1.1.50: bytes=32 time=10ms TTL=124
Reply from 10.1.1.50: bytes=32 time=4ms TTL=124
Reply from 10.1.1.50: bytes=32 time=10ms TTL=124
Reply from 10.1.1.50: bytes=32 time=10ms TTL=124

Ping statistics for 10.1.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 10ms, Average = 8ms

C:\>ping 10.4.1.50

Pinging 10.4.1.50 with 32 bytes of data:

Reply from 10.4.1.50: bytes=32 time=4ms TTL=125
Reply from 10.4.1.50: bytes=32 time=4ms TTL=125
Reply from 10.4.1.50: bytes=32 time=4ms TTL=125
Reply from 10.4.1.50: bytes=32 time=3ms TTL=125

Ping statistics for 10.4.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 4ms, Average = 3ms

C:\>
```

15- Ancash

- 1 Router 1941
- 1 Switch 2960
- 1 Server-PT
- 1 PC-PT

Y se conectaran con cables **straight-through**.

Cable SERIAL (rojo)

- Huanaco Serial 0/0/1 → Ancash Serial 0/0/0

Configuracion de Router

```
enable
configure terminal

interface serial 0/0/0
ip address 172.19.1.2 255.255.255.252
no shutdown

interface gigabitethernet 0/0
ip address 10.2.1.1 255.255.255.0
no shutdown

ip route 0.0.0.0 0.0.0.0 172.19.1.1

end
write memory
```


Actualizar router Huanuco

```
enable
configure terminal

interface serial 0/0/1
ip address 172.19.1.1 255.255.255.252
clock rate 64000
no shutdown

ip route 10.2.0.0 255.255.0.0 172.19.1.2

end
write memory
```

Configuracion de switch

```
enable
configure terminal

interface vlan 1
ip address 10.2.1.10 255.255.255.0
no shutdown
exit

ip default-gateway 10.2.1.1

end
write memory
```

Configuracion del server

- IP Address: 10.2.1.100
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.2.1.1

Configuracion de PC

- IP Address: 10.2.1.50
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.2.1.1

Pruebas de conectividad

Command Prompt

Cisco Packet Tracer PC Command Line 1.0

C:\>ping 10.1.1.50

Pinging 10.1.1.50 with 32 bytes of data:

Reply from 10.1.1.50: bytes=32 time=10ms TTL=123

Reply from 10.1.1.50: bytes=32 time=6ms TTL=123

Reply from 10.1.1.50: bytes=32 time=5ms TTL=123

Reply from 10.1.1.50: bytes=32 time=71ms TTL=123

Ping statistics for 10.1.1.50:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 5ms, Maximum = 71ms, Average = 23ms

C:\>ping 10.4.1.50

Pinging 10.4.1.50 with 32 bytes of data:

Reply from 10.4.1.50: bytes=32 time=33ms TTL=124

Reply from 10.4.1.50: bytes=32 time=3ms TTL=124

Reply from 10.4.1.50: bytes=32 time=4ms TTL=124

Reply from 10.4.1.50: bytes=32 time=34ms TTL=124

Ping statistics for 10.4.1.50:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 3ms, Maximum = 34ms, Average = 18ms

C:\>ping 10.18.1.50

Pinging 10.18.1.50 with 32 bytes of data:

Reply from 10.18.1.50: bytes=32 time=32ms TTL=123

Reply from 10.18.1.50: bytes=32 time=5ms TTL=123

Reply from 10.18.1.50: bytes=32 time=31ms TTL=123

Reply from 10.18.1.50: bytes=32 time=4ms TTL=123

Ping statistics for 10.18.1.50:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 4ms, Maximum = 32ms, Average = 18ms

16- Pasco

- 1 Router 1941
- 1 Switch 2960
- 1 Server-PT
- 1 PC-PT

Y se conectaran con cables **straight-through**.

Cable SERIAL (rojo)

- Huanaco Serial 0/0/1 → Ancash Serial 0/0/0

Configuracion de Router

```
enable
configure terminal

interface serial 0/0/0
ip address 172.19.2.2 255.255.255.252
no shutdown

interface gigabitethernet 0/0
ip address 10.19.1.1 255.255.255.0
no shutdown

ip route 0.0.0.0 0.0.0.0 172.19.2.1

end
write memory
```

Actualizar router Huanuco

```
enable
configure terminal

interface serial 0/2/0
ip address 172.19.2.1 255.255.255.252
clock rate 64000
no shutdown

ip route 10.19.0.0 255.255.0.0 172.19.2.2

end
write memory
```

Configuracion de switch

```
enable
configure terminal

interface vlan 1
ip address 10.19.1.10 255.255.255.0
no shutdown
exit

ip default-gateway 10.19.1.1

end
write memory
```

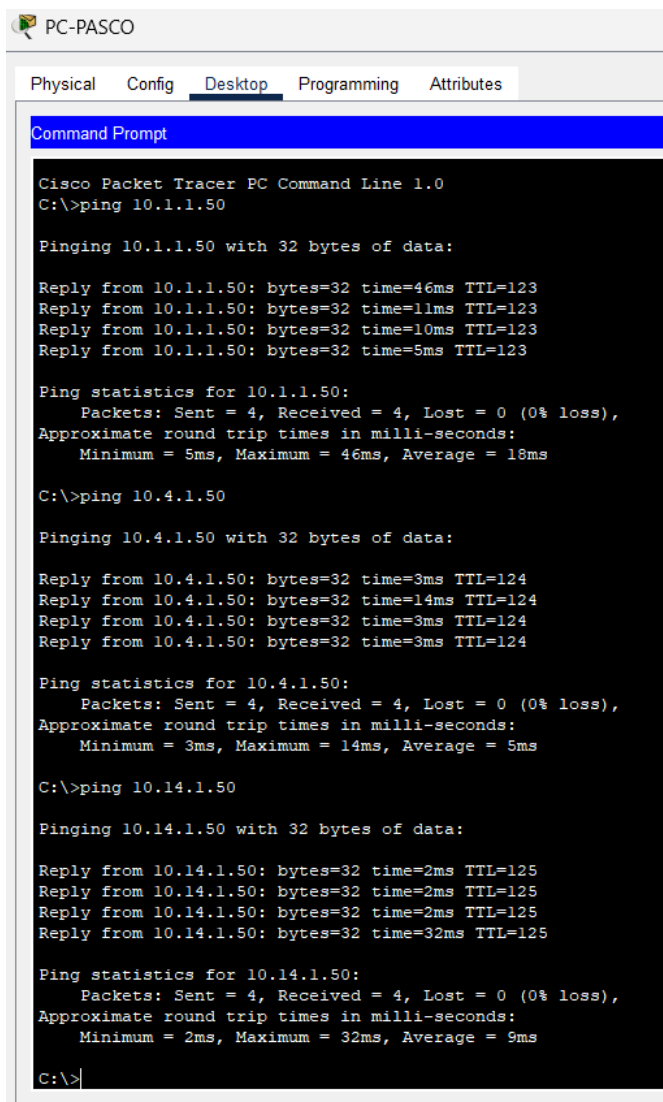
Configuracion del server

- IP Address: 10.19.1.100
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.19.1.1

Configuracion de PC

- IP Address: 10.19.1.50
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.19.1.1

Pruebas de conectividad



The screenshot shows a PC-PASCO interface with a 'Desktop' tab selected. A 'Command Prompt' window is open, displaying the results of three ping tests performed from the C:\> prompt. The first test is to 10.1.1.50, the second to 10.4.1.50, and the third to 10.14.1.50. Each test shows four successful replies with varying round-trip times and TTL values, and statistics indicating 0% packet loss.

```
PC-PASCO
Physical  Config  Desktop  Programming  Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.1.1.50

Pinging 10.1.1.50 with 32 bytes of data:

Reply from 10.1.1.50: bytes=32 time=46ms TTL=123
Reply from 10.1.1.50: bytes=32 time=11ms TTL=123
Reply from 10.1.1.50: bytes=32 time=10ms TTL=123
Reply from 10.1.1.50: bytes=32 time=5ms TTL=123

Ping statistics for 10.1.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 46ms, Average = 18ms

C:\>ping 10.4.1.50

Pinging 10.4.1.50 with 32 bytes of data:

Reply from 10.4.1.50: bytes=32 time=3ms TTL=124
Reply from 10.4.1.50: bytes=32 time=14ms TTL=124
Reply from 10.4.1.50: bytes=32 time=3ms TTL=124
Reply from 10.4.1.50: bytes=32 time=3ms TTL=124

Ping statistics for 10.4.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 14ms, Average = 5ms

C:\>ping 10.14.1.50

Pinging 10.14.1.50 with 32 bytes of data:

Reply from 10.14.1.50: bytes=32 time=2ms TTL=125
Reply from 10.14.1.50: bytes=32 time=2ms TTL=125
Reply from 10.14.1.50: bytes=32 time=2ms TTL=125
Reply from 10.14.1.50: bytes=32 time=32ms TTL=125

Ping statistics for 10.14.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 32ms, Average = 9ms

C:\>|
```

17- Junín

- 1 Router 1941
- 1 Switch 2960
- 1 Server-PT
- 1 PC-PT

Y se conectaran con cables **straight-through**.

Cable SERIAL (rojo)

- Huánuco Serial 0/1/0 → Junín Serial 0/0/0

Configuracion de Router

```
enable
configure terminal

interface serial 0/0/0
ip address 172.19.3.2 255.255.255.252
no shutdown

interface gigabitethernet 0/0
ip address 10.12.1.1 255.255.255.0
no shutdown

ip route 0.0.0.0 0.0.0.0 172.19.3.1

end
write memory
```

Actualizar router Huanuco

```
enable
configure terminal

interface serial 0/1/0
ip address 172.19.3.1 255.255.255.252
clock rate 64000
no shutdown

ip route 10.12.0.0 255.255.0.0 172.19.3.2

end
write memory
```

Configuracion de switch

```
enable
configure terminal

interface vlan 1
ip address 10.12.1.10 255.255.255.0
no shutdown
exit

ip default-gateway 10.12.1.1

end
write memory
```

Configuracion del server

- IP Address: 10.12.1.100
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.12.1.1

Configuracion de PC

- IP Address: 10.12.1.50
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.12.1.1

Pruebas de conectividad

Command Prompt

Cisco Packet Tracer PC Command Line 1.0

C:\>ping 10.2.1.50

Pinging 10.2.1.50 with 32 bytes of data:

Reply from 10.2.1.50: bytes=32 time=2ms TTL=125

Reply from 10.2.1.50: bytes=32 time=2ms TTL=125

Reply from 10.2.1.50: bytes=32 time=2ms TTL=125

Reply from 10.2.1.50: bytes=32 time=2ms TTL=125

Ping statistics for 10.2.1.50:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:

Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\>ping 10.1.1.50

Pinging 10.1.1.50 with 32 bytes of data:

Reply from 10.1.1.50: bytes=32 time=64ms TTL=123

Reply from 10.1.1.50: bytes=32 time=48ms TTL=123

Reply from 10.1.1.50: bytes=32 time=29ms TTL=123

Reply from 10.1.1.50: bytes=32 time=19ms TTL=123

Ping statistics for 10.1.1.50:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:

Minimum = 19ms, Maximum = 64ms, Average = 40ms

C:\>ping 10.4.1.50

Pinging 10.4.1.50 with 32 bytes of data:

Reply from 10.4.1.50: bytes=32 time=3ms TTL=124

Reply from 10.4.1.50: bytes=32 time=3ms TTL=124

Reply from 10.4.1.50: bytes=32 time=3ms TTL=124

Reply from 10.4.1.50: bytes=32 time=3ms TTL=124

Ping statistics for 10.4.1.50:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:

Minimum = 3ms, Maximum = 3ms, Average = 3ms

C:\>

18- Huancavelica

- 1 Router 1941
- 1 Switch 2960
- 1 Server-PT
- 1 PC-PT

Y se conectaran con cables **straight-through**.

Cable SERIAL (rojo)

- Huánuco Serial 0/1/0 → Junín Serial 0/0/0

Configuracion de Router

```
enable
configure terminal

interface serial 0/0/0
ip address 172.19.4.2 255.255.255.252
no shutdown

interface gigabitethernet 0/0
ip address 10.9.1.1 255.255.255.0
no shutdown

ip route 0.0.0.0 0.0.0.0 172.19.4.1

end
write memory
```

Actualizar router Huanuco

```
enable
configure terminal

interface serial 0/1/1
ip address 172.19.4.1 255.255.255.252
clock rate 64000
no shutdown

ip route 10.9.0.0 255.255.0.0 172.19.4.2

end
write memory
```

Configuracion de switch

```
enable
configure terminal

interface vlan 1
ip address 10.9.1.10 255.255.255.0
no shutdown
exit

ip default-gateway 10.9.1.1

end
write memory
```

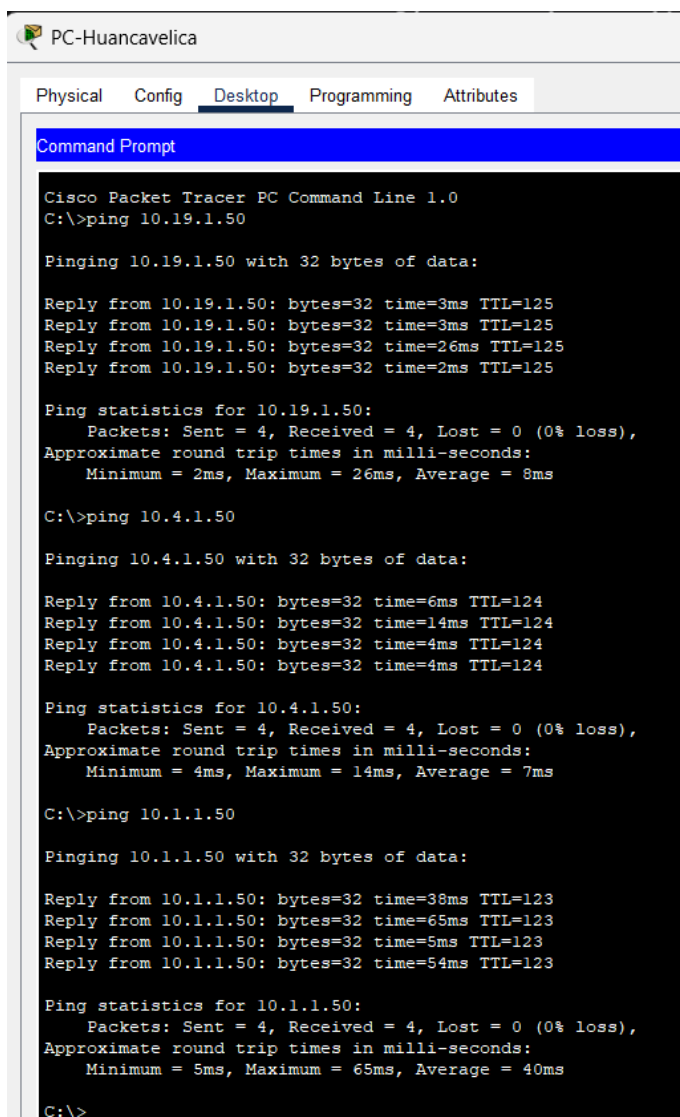
Configuracion del server

- IP Address: 10.9.1.100
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.9.1.1

Configuracion de PC

- IP Address: 10.9.1.50
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.9.1.1

Pruebas de conectividad



```
PC-Huancavelica
Physical  Config  Desktop  Programming  Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.19.1.50

Pinging 10.19.1.50 with 32 bytes of data:

Reply from 10.19.1.50: bytes=32 time=3ms TTL=125
Reply from 10.19.1.50: bytes=32 time=3ms TTL=125
Reply from 10.19.1.50: bytes=32 time=26ms TTL=125
Reply from 10.19.1.50: bytes=32 time=2ms TTL=125

Ping statistics for 10.19.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 26ms, Average = 8ms

C:\>ping 10.4.1.50

Pinging 10.4.1.50 with 32 bytes of data:

Reply from 10.4.1.50: bytes=32 time=6ms TTL=124
Reply from 10.4.1.50: bytes=32 time=14ms TTL=124
Reply from 10.4.1.50: bytes=32 time=4ms TTL=124
Reply from 10.4.1.50: bytes=32 time=4ms TTL=124

Ping statistics for 10.4.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 14ms, Average = 7ms

C:\>ping 10.1.1.50

Pinging 10.1.1.50 with 32 bytes of data:

Reply from 10.1.1.50: bytes=32 time=38ms TTL=123
Reply from 10.1.1.50: bytes=32 time=65ms TTL=123
Reply from 10.1.1.50: bytes=32 time=5ms TTL=123
Reply from 10.1.1.50: bytes=32 time=54ms TTL=123

Ping statistics for 10.1.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 65ms, Average = 40ms

C:\>
```

SERVIDORES REGIONALES

Region Centro-Oriente

19- LORETO

- 1 Router 2911
- 1 Switch 2960
- 1 Server-PT
- 1 PC-PT

Y se conectaran con cables **straight-through**.

Cable SERIAL (rojo)

Lima Serial 0/2/0 → **LORETO** Serial 0/0/0

Configuracion de Router

```
enable
configure terminal

interface serial 0/0/0
ip address 172.16.4.2 255.255.255.252
no shutdown

interface gigabitethernet 0/0
ip address 10.16.1.1 255.255.255.0

no shutdown

ip route 0.0.0.0 0.0.0.0 172.16.4.1
end
write memory
```

Actualizar router Lima

```
enable
configure terminal
interface serial 0/2/0
ip address 172.16.4.1 255.255.255.252
clock rate 64000
no shutdown
ip route 10.16.0.0 255.255.0.0 172.16.4.2
end
write memory
```

Configuracion de switch

```
enable
configure terminal
interface vlan 1
ip address 10.16.1.10 255.255.255.0
no shutdown
exit
ip default-gateway 10.16.1.1
end
write memory
```

Configuracion del server

- IP Address: 10.16.1.100
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.16.1.1

Configuracion de PC

- IP Address: 10.16.1.50
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.16.1.1

Pruebas de conectividad

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.14.1.1

Pinging 10.14.1.1 with 32 bytes of data:

Reply from 10.14.1.1: bytes=32 time=2ms TTL=254
Reply from 10.14.1.1: bytes=32 time=1ms TTL=254
Reply from 10.14.1.1: bytes=32 time=34ms TTL=254
Reply from 10.14.1.1: bytes=32 time=1ms TTL=254

Ping statistics for 10.14.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 34ms, Average = 9ms

C:\>ping 10.14.1.100

Pinging 10.14.1.100 with 32 bytes of data:

Request timed out.
Reply from 10.14.1.100: bytes=32 time=1ms TTL=126
Reply from 10.14.1.100: bytes=32 time=3ms TTL=126
Reply from 10.14.1.100: bytes=32 time=1ms TTL=126

Ping statistics for 10.14.1.100:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 1ms

C:\>ping 10.14.1.50

Pinging 10.14.1.50 with 32 bytes of data:

Request timed out.
Reply from 10.14.1.50: bytes=32 time=21ms TTL=126
Reply from 10.14.1.50: bytes=32 time=20ms TTL=126
Reply from 10.14.1.50: bytes=32 time=1ms TTL=126

Ping statistics for 10.14.1.50:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 21ms, Average = 14ms
```

20- San Martin

- 1 Router 1941
- 1 Switch 2960
- 1 Server-PT
- 1 PC-PT

Y se conectaran con cables **straight-through**.

Cable SERIAL (rojo)

LORETOs Serial 0/0/1 → **San Martin** Serial 0/0/0

Configuracion de Router

```
enable
configure terminal
interface serial 0/0/0
ip address 172.20.1.2 255.255.255.252
no shutdown
interface gigabitethernet 0/0
ip address 10.15.1.1 255.255.255.0
no shutdown
ip route 0.0.0.0 0.0.0.0 172.20.1.1
end
write memory
```

Actualizar router LORETOs

```
enable
configure terminal
interface serial 0/0/1
ip address 172.20.1.1 255.255.255.252
clock rate 64000
no shutdown
ip route 10.15.0.0 255.255.0.0 172.20.1.2
end
write memory
```

Configuracion de switch

```
enable
configure terminal
hostname SW-SANMARTIN
interface vlan 1
ip address 10.15.1.10 255.255.255.0
no shutdown
exit
ip default-gateway 10.15.1.1
end
write memory
```

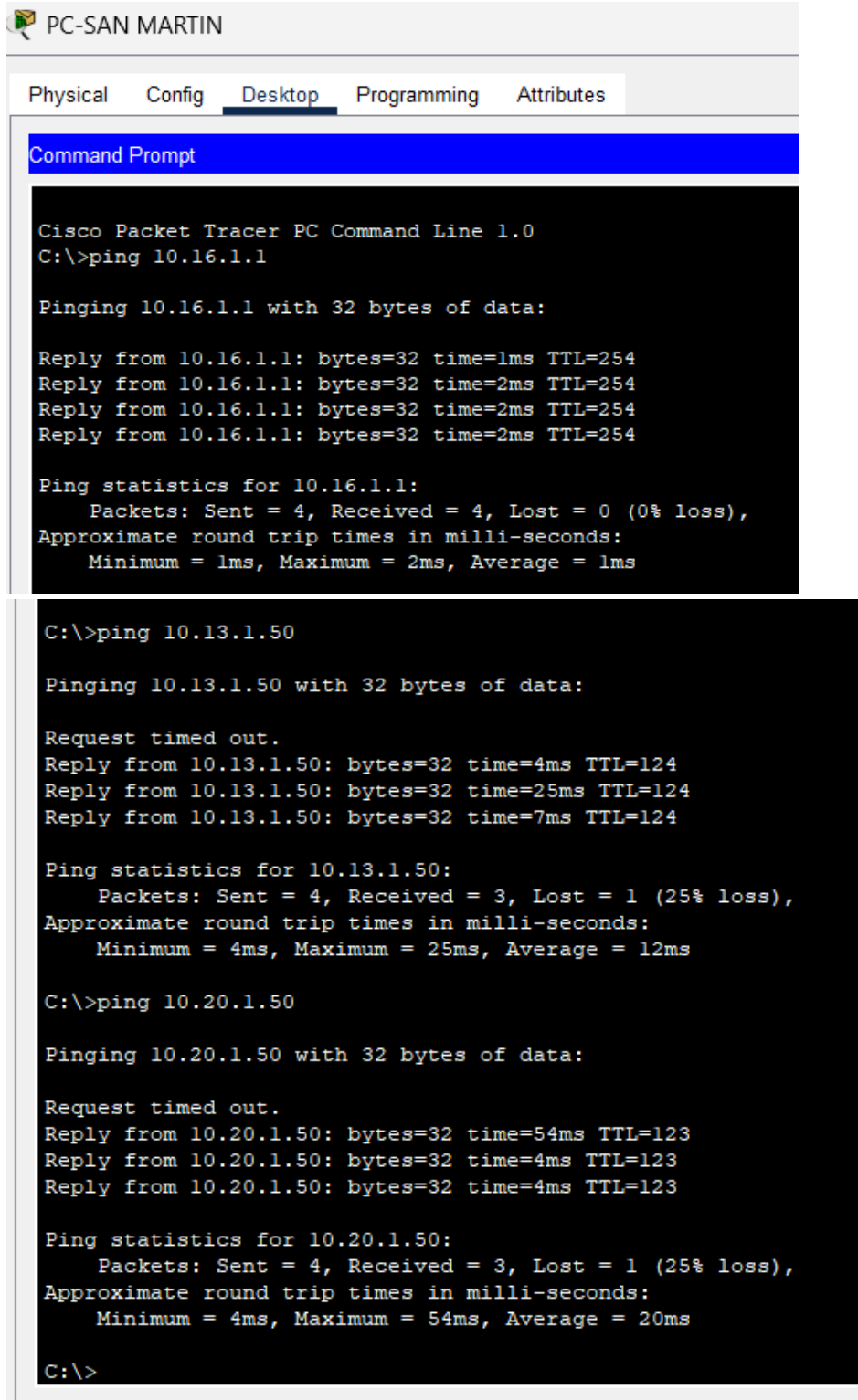
Configuracion del server

- IP Address: 10.15.1.100
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.15.1.1

Configuracion de PC

- IP Address: 10.15.1.50
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.15.1.1

Pruebas de conectividad



The screenshot shows the Cisco Packet Tracer PC Command Line interface for a PC named 'PC-SAN MARTIN'. The 'Desktop' tab is selected. The Command Prompt window displays the results of two ping tests. The first test is to 10.16.1.1, which is successful with 0% loss. The second test is to 10.13.1.50, which fails with 25% loss. The third test is to 10.20.1.50, which also fails with 25% loss.

```
PC-SAN MARTIN
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.16.1.1

Pinging 10.16.1.1 with 32 bytes of data:

Reply from 10.16.1.1: bytes=32 time=1ms TTL=254
Reply from 10.16.1.1: bytes=32 time=2ms TTL=254
Reply from 10.16.1.1: bytes=32 time=2ms TTL=254
Reply from 10.16.1.1: bytes=32 time=2ms TTL=254

Ping statistics for 10.16.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>ping 10.13.1.50

Pinging 10.13.1.50 with 32 bytes of data:

Request timed out.
Reply from 10.13.1.50: bytes=32 time=4ms TTL=124
Reply from 10.13.1.50: bytes=32 time=25ms TTL=124
Reply from 10.13.1.50: bytes=32 time=7ms TTL=124

Ping statistics for 10.13.1.50:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 25ms, Average = 12ms

C:\>ping 10.20.1.50

Pinging 10.20.1.50 with 32 bytes of data:

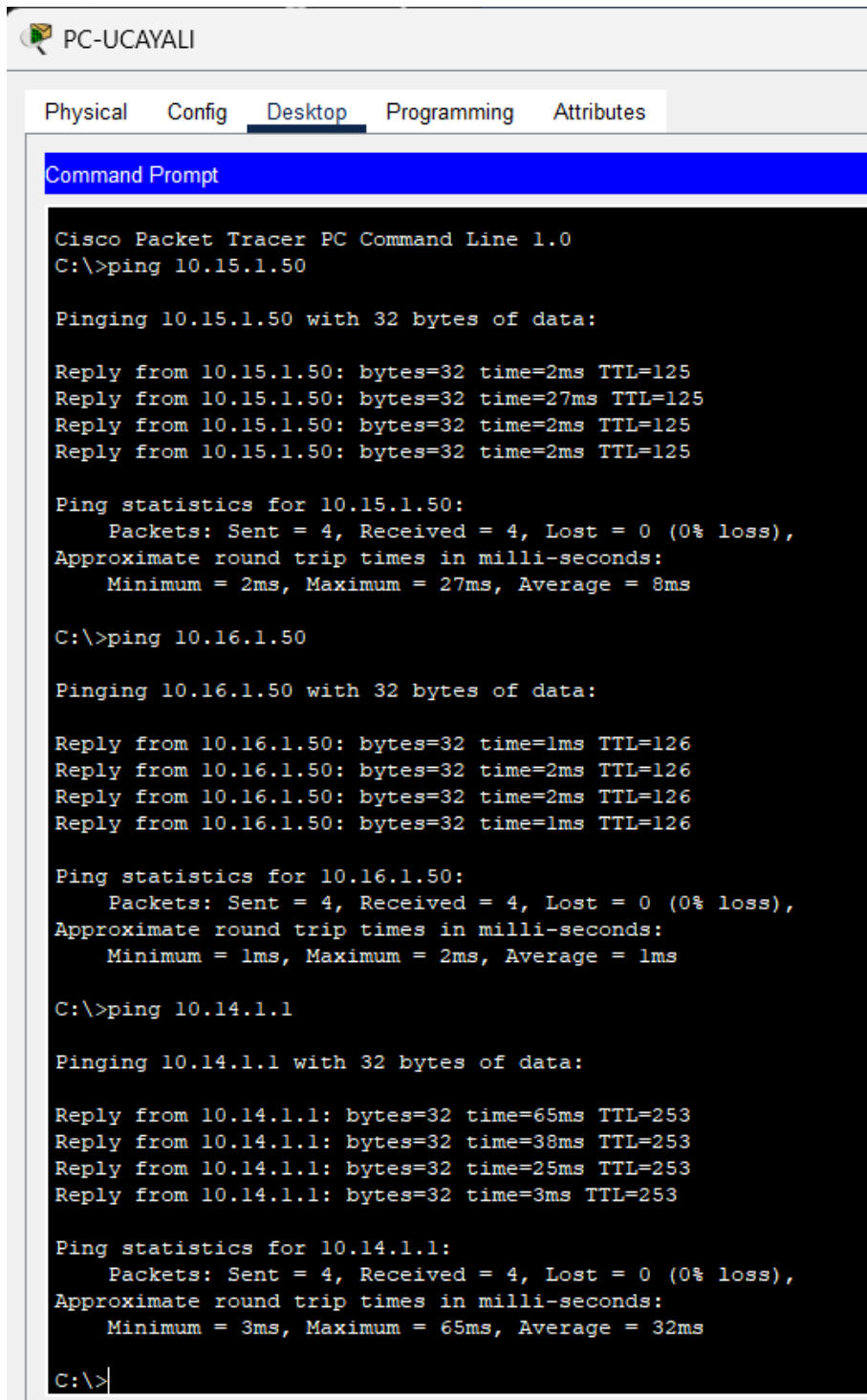
Request timed out.
Reply from 10.20.1.50: bytes=32 time=54ms TTL=123
Reply from 10.20.1.50: bytes=32 time=4ms TTL=123
Reply from 10.20.1.50: bytes=32 time=4ms TTL=123

Ping statistics for 10.20.1.50:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 54ms, Average = 20ms

C:\>
```

Ya para no alargarme mas paginas solo mostrare conectividades desde ahorita y sus estados

21- UCAYALI



```
PC-UCAYALI
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.15.1.50

Pinging 10.15.1.50 with 32 bytes of data:

Reply from 10.15.1.50: bytes=32 time=2ms TTL=125
Reply from 10.15.1.50: bytes=32 time=27ms TTL=125
Reply from 10.15.1.50: bytes=32 time=2ms TTL=125
Reply from 10.15.1.50: bytes=32 time=2ms TTL=125

Ping statistics for 10.15.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 27ms, Average = 8ms

C:\>ping 10.16.1.50

Pinging 10.16.1.50 with 32 bytes of data:

Reply from 10.16.1.50: bytes=32 time=1ms TTL=126
Reply from 10.16.1.50: bytes=32 time=2ms TTL=126
Reply from 10.16.1.50: bytes=32 time=2ms TTL=126
Reply from 10.16.1.50: bytes=32 time=1ms TTL=126

Ping statistics for 10.16.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>ping 10.14.1.1

Pinging 10.14.1.1 with 32 bytes of data:

Reply from 10.14.1.1: bytes=32 time=65ms TTL=253
Reply from 10.14.1.1: bytes=32 time=38ms TTL=253
Reply from 10.14.1.1: bytes=32 time=25ms TTL=253
Reply from 10.14.1.1: bytes=32 time=3ms TTL=253

Ping statistics for 10.14.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 65ms, Average = 32ms

C:\>|
```

22- MADRE DE DIOS



PC-MADRE DE DIOS

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.16.1.100

Pinging 10.16.1.100 with 32 bytes of data:

Request timed out.
Reply from 10.16.1.100: bytes=32 time=21ms TTL=126
Reply from 10.16.1.100: bytes=32 time=1ms TTL=126
Reply from 10.16.1.100: bytes=32 time=2ms TTL=126

Ping statistics for 10.16.1.100:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 21ms, Average = 8ms

C:\>ping 10.15.1.50

Pinging 10.15.1.50 with 32 bytes of data:

Reply from 10.15.1.50: bytes=32 time=2ms TTL=125
Reply from 10.15.1.50: bytes=32 time=23ms TTL=125
Reply from 10.15.1.50: bytes=32 time=36ms TTL=125
Reply from 10.15.1.50: bytes=32 time=2ms TTL=125

Ping statistics for 10.15.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 36ms, Average = 15ms

C:\>ping 10.25.1.50

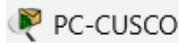
Pinging 10.25.1.50 with 32 bytes of data:

Reply from 10.25.1.50: bytes=32 time=28ms TTL=125
Reply from 10.25.1.50: bytes=32 time=2ms TTL=125
Reply from 10.25.1.50: bytes=32 time=2ms TTL=125
Reply from 10.25.1.50: bytes=32 time=2ms TTL=125

Ping statistics for 10.25.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 28ms, Average = 8ms

C:\>|
```

23- CUSCO



Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.21.1.50

Pinging 10.21.1.50 with 32 bytes of data:

Reply from 10.21.1.50: bytes=32 time=3ms TTL=125
Reply from 10.21.1.50: bytes=32 time=3ms TTL=125
Reply from 10.21.1.50: bytes=32 time=5ms TTL=125
Reply from 10.21.1.50: bytes=32 time=24ms TTL=125

Ping statistics for 10.21.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 24ms, Average = 8ms

C:\>ping 10.25.1.50

Pinging 10.25.1.50 with 32 bytes of data:

Reply from 10.25.1.50: bytes=32 time=41ms TTL=125
Reply from 10.25.1.50: bytes=32 time=2ms TTL=125
Reply from 10.25.1.50: bytes=32 time=2ms TTL=125
Reply from 10.25.1.50: bytes=32 time=2ms TTL=125

Ping statistics for 10.25.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 41ms, Average = 11ms

C:\>ping 10.15.1.50

Pinging 10.15.1.50 with 32 bytes of data:

Reply from 10.15.1.50: bytes=32 time=30ms TTL=125
Reply from 10.15.1.50: bytes=32 time=3ms TTL=125
Reply from 10.15.1.50: bytes=32 time=2ms TTL=125
Reply from 10.15.1.50: bytes=32 time=2ms TTL=125

Ping statistics for 10.15.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 30ms, Average = 9ms

C:\>
```

24- PUNO



PC-PUNO

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.8.1.50

Pinging 10.8.1.50 with 32 bytes of data:

Reply from 10.8.1.50: bytes=32 time=2ms TTL=125
Reply from 10.8.1.50: bytes=32 time=39ms TTL=125
Reply from 10.8.1.50: bytes=32 time=2ms TTL=125
Reply from 10.8.1.50: bytes=32 time=2ms TTL=125

Ping statistics for 10.8.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 39ms, Average = 11ms

C:\>ping 10.15.1.50

Pinging 10.15.1.50 with 32 bytes of data:

Reply from 10.15.1.50: bytes=32 time=39ms TTL=125
Reply from 10.15.1.50: bytes=32 time=44ms TTL=125
Reply from 10.15.1.50: bytes=32 time=2ms TTL=125
Reply from 10.15.1.50: bytes=32 time=2ms TTL=125

Ping statistics for 10.15.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 44ms, Average = 21ms

C:\>ping 10.14.1.100

Pinging 10.14.1.100 with 32 bytes of data:

Reply from 10.14.1.100: bytes=32 time=28ms TTL=125
Reply from 10.14.1.100: bytes=32 time=38ms TTL=125
Reply from 10.14.1.100: bytes=32 time=33ms TTL=125
Reply from 10.14.1.100: bytes=32 time=2ms TTL=125

Ping statistics for 10.14.1.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 38ms, Average = 25ms

C:\>
```


SERVIDOR NACIONAL(VISUALMENTE)

